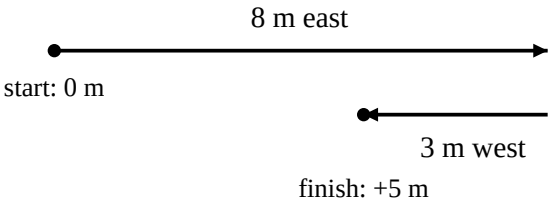
Describing Motion

1. Distance and displacement

Distance: the whole path. **Displacement:** finish - start.



Distance = 8 + 3 = **11 m**
Displacement = 5 - 0 = **5 m east**

2. Speed and velocity

$$\text{average speed} = \frac{\text{total distance}}{\text{total elapsed time}}$$

Example: 150 m in 100 s
150 ÷ 100 = **1.5 m/s**

Distance = speed × time **Time = distance ÷ speed**

2 m/s east

2 m/s west

Same speed. Different velocities because direction is different.

3. Acceleration: a change in velocity

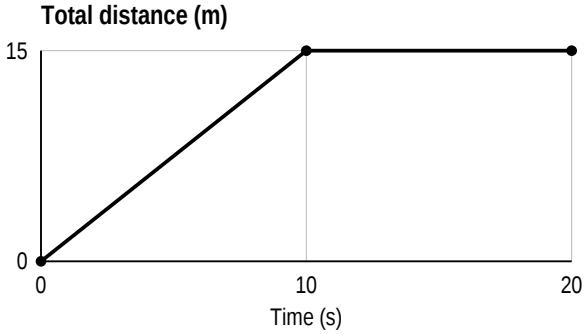
$$a = \frac{\text{final velocity - initial velocity}}{\text{elapsed time}}$$

Elapsed time = final time - initial time.
From 2 s to 6 s: 6 - 2 = 4 s.

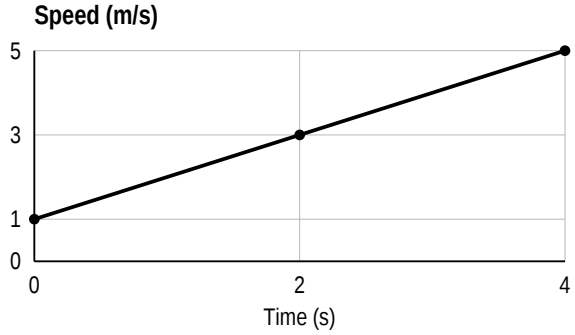
Example: a helicopter rises from 3 m/s to 9 m/s in 4 s. Take upward as positive.
1. Find the change: 9 - 3 = 6 m/s 2. Divide: 6 ÷ 4 = 1.5 m/s² upward

v_i = initial velocity; v_f = final velocity. $\mathbf{v_f = v_i + a \times}$ **elapsed time.**
Acceleration opposite velocity slows an object. A minus sign shows direction, not always slowing.

4. Read the axes before reading the graph



Slope = speed. Flat means stopped.
First 10 s: 15 ÷ 10 = **1.5 m/s**.



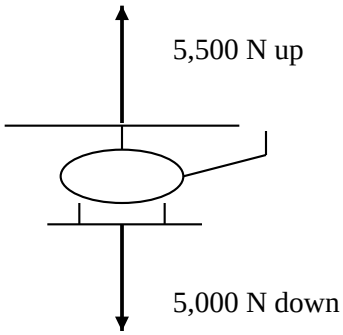
For straight motion in the positive direction:
(5 - 1) ÷ 4 = **1 m/s²**.

A speed graph does not show direction. Constant speed can still include turning.

The Sun's gravity keeps a planet's path curved.

Vertical Forces, Gravity, and Collisions

1. Find weight, net force, then acceleration



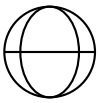
500 kg helicopter

Take upward as positive. Use g = 10 m/s².
Ignore other vertical forces.
1. Weight: 500 × 10 = **5,000 N down**.
2. Net force: 5,500 - 5,000 = **+500 N**.
3. Acceleration: 500 ÷ 500 = **1 m/s² up**.

Weight = m × g; $\mathbf{F_{net} = ma;}$ $\mathbf{a = F_{net} \div m;}$ $\mathbf{m = F_{net} \div a}$

Net force sets acceleration, not the current direction of travel.
Zero net force: rest or constant velocity. More mass means more inertia.

2. Same mass, different planet, different weight



Earth (g = 10 m/s²)
3 kg × 10 = **30 N**

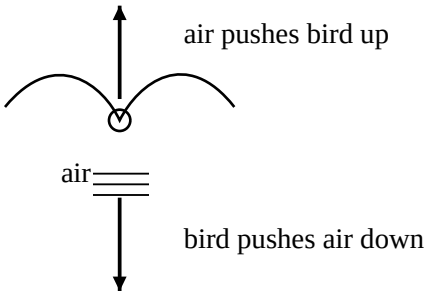


Mars (g = 3.7 m/s²)
3 kg × 3.7 = **11.1 N**

The object keeps its **3 kg mass**. Its **weight** changes because gravity is different.

Friction opposes slipping; static friction adjusts up to its maximum.

3. A bird and the air: forces on different objects

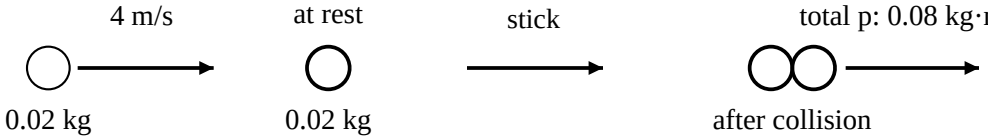


Equal size. Opposite directions.
One force acts on the **bird**.
The other acts on the **air**.
They do not cancel on the bird.

4. Total momentum of a two-object system

p = mv; $\mathbf{p_{total} = m_1v_1 + m_2v_2}$

Momentum unit: kg·m/s



Ignore external horizontal forces during the brief collision.

1. Each piece: 0.02 × 4 = **0.08 kg·m/s**; 0.02 × 0 = **0 kg·m/s**.
2. Add the momenta: 0.08 + 0 = **0.08 kg·m/s right**.
3. After the collision, total momentum is still **0.08 kg·m/s right**.